

Plan Project n°1: *Bert has uncommon sense*, Luke Gessler

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1 Summary of the Paper

For our project, we focus on the paper [1], which examines how well contextualized word embedding models like BERT capture different word senses, particularly rare ones. Instead of using a traditional word sense disambiguation (WSD) approach, the authors propose a similarity ranking method based on cosine similarity between word embeddings in context. Performance is evaluated using Mean Average Precision (MAP) on the top 50 results. The study tests several pre-trained models—BERT, RoBERTa, DistilBERT, ALBERT, XLNet, and GPT-2—on two English sense-annotated corpora: OntoNotes 5.0 (covering nouns and verbs) and PDEP (Pattern Dictionary of English Prepositions). Results show that while all models outperform random ranking, they vary significantly, especially for rare senses. BERT achieves the best overall performance, while RoBERTa, despite excelling on other benchmarks, struggles with rare sense representation. The study reveals key differences between similar models and questions whether standard NLP benchmarks truly reflect their ability to capture lexical semantics.

2 Project Plan

2.1 Reproducing the paper's results

For this project, we will first aim to reproduce the results of the paper [1], relying on its original code [2]. This involves comparing the performance of BERT, RoBERTa, DistilBERT, ALBERT, XLNet, and GPT-2 on the OntoNotes 5.0 and PDEP datasets across different levels of fine-tuning (0, 100, 250, 500, 1000, 2500), within the limits of our computational resources.

Once we have reproduced the paper's results, we propose a list of ideas to further extend the analysis. However, as these approaches remain untested and the required time is uncertain, we may not be able to implement all of them.

2.2 Testing new datasets and models

We plan to extend the experiments by testing the code on a new French-language model and dataset. Specifically, we will evaluate CamemBERT-cased and, if possible, FlauBERT, using the FLUE dataset for word sense disambiguation (FLUE-WSD) [3].

Next, we will test the FLUE dataset across all other models. Since CamemBERT is specifically designed for French, we expect it to perform better than the other models.

Finally, we could also evaluate models such as BART and T5 on the English datasets OntoNotes 5.0 and PDEP, which were not tested in the original paper. We would then compare their results with those of the previously tested models to assess their effectiveness in word sense representation.

2.3 Testing new metrics

In the original paper, word embeddings are compared using cosine similarity: a high cosine similarity means very similar embeddings, while a low one indicates dissimilarity. However, concerns have been raised, es-

pecially as noted in [4], about the reliability of this metric, particularly when dealing with very frequent words.

In this part, we want to test how robust our results are to a change of metric. To do so, we will first rerun the experiments using well-known alternatives, namely the Euclidean distance and the Jaccard index, and check whether we obtain similar results to those obtained with cosine similarity.

Beyond that, we also want to explore other ways of identifying examples in the dataset that share the same meaning as our target word. Following [5], we will implement classification-based methods, such as thresholding, to evaluate the relevance of CWE for word sense disambiguation (WSD).

2.4 Exploring the Embedding Layers

The original paper systematically uses the output representations of the final layer of each model to compute the similarities between embeddings. The underlying rationale for this choice is that the final layer is expected to encapsulate the most informative and contextually rich representation of each vector. This assumption is based on the fact that, after passing through multiple layers of self-attention, the model refines its understanding of linguistic subtleties, leading to a more meaningful embedding.

In this section, we would like to empirically evaluate this assumption by assessing whether the final layer indeed provides the most effective representation. To this end, we will first replicate the original experiments using the embeddings from the first layer instead of the final one and compare the results. Since the first layer has undergone minimal transformation of the input data, we should see a decline in performance. If the results from the first layer prove significantly worse, we will incrementally test deeper layers to determine the point at which the model begins to produce high-quality representations. This analysis will provide insights into how representational quality evolves across different layers of the model. It will also help us to understand the importance of the depth (in terms of layers) of these models.

References

1. Gessler, L. & Schneider, N. *BERT Has Uncommon Sense: Similarity Ranking for Word Sense BERTology* in *Proceedings of the Fourth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP* (eds Bastings, J. et al.) (Association for Computational Linguistics, Punta Cana, Dominican Republic, Nov. 2021), 539–547. <https://aclanthology.org/2021.blackboxnlp-1.43/>.
2. Gessler, L. *BERT Has Uncommon Sense - GitHub Repository* Accessed: February 2025. 2021. <https://github.com/lgessler/bert-has-uncommon-sense>.
3. Le, H. et al. *FlauBERT: Unsupervised Language Model Pre-training for French* Accessed: February 2025. 2019. arXiv: 1912.05372 [cs.CL]. <https://github.com/getalp/Flaubert/tree/master/flue>.
4. Zhou, K., Ethayarajh, K., Card, D. & Jurafsky, D. *Problems with Cosine as a Measure of Embedding Similarity for High Frequency Words* in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (eds Muresan, S., Nakov, P. & Villavicencio, A.) (Association for Computational Linguistics, Dublin, Ireland, May 2022), 401–423. <https://aclanthology.org/2022.acl-short.45/>.
5. Pilehvar, M. T. & Camacho-Collados, J. *WiC: the Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations* in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (eds Burstein, J., Doran, C. & Solorio, T.) (Association for Computational Linguistics, Minneapolis, Minnesota, June 2019), 1267–1273. <https://aclanthology.org/N19-1128/>.